COMPUTER VISION PROJECT SUMMARY

Classification, Segmentation & Adversarial Attacks on PASCAL VOC

Natalia Iglesias Bueno | Group project (KU Leuven)

PROJECT DETAILS

- ***Role*** → Computer Vision & Deep Learning Engineer (group of four)
- ***Setting*** → KU Leuven, course H02A5a Computer Vision, Group assignment
- ***Technical Stack*** →
  - Python, PyTorch, torchvision, scikit-learn
  - Models: timm (ConvNeXt, ViT), DINOv2 (Hugging Face), segmentation_models_pytorch
  - Data and training: albumentations, iterative-stratification, mixed precision
- ***Core Domains*** →
  - Multi-label image classification and transfer learning
  - Semantic segmentation with a frozen self-supervised backbone (DINOv2)
  - Loss design for class imbalance (Asymmetric Loss, Dice + Cross-Entropy)
  - Adversarial attacks (PGD) and explainability (Grad-CAM)

PROBLEM STATEMENT

The project uses a subset of **PASCAL VOC**: **749 labelled training images** and 750 test images, with 20 object classes. Models had to solve two tasks on every image, and then be tested for robustness:

- **Classification (multi-label):** predict which of the 20 classes appear in the image. Several classes can appear together, so each class needs its own yes/no output.
- **Segmentation:** give every pixel one of 21 labels (20 classes plus background), roughly 200,000 predictions per image.
- **Adversarial attacks:** change an image slightly, so that the trained models make a wrong prediction.

The main difficulty is the small dataset, together with strong class imbalance: some classes have very few examples, and background covers about 70–80% of the pixels.

The objective

🎯 ***Main goal*** → *find the best way to train classification and segmentation models on a small dataset, and test how robust they are to attacks.*

Steps required:

First objective → *classification: compare CNNs trained from scratch with transfer learning from ImageNet-21k.*

Second objective → *segmentation: compare U-Net, DeepLabV3+ and a DINOv2-based model.*

Third objective → *attack the best models with PGD and use Grad-CAM to explain what happens.*

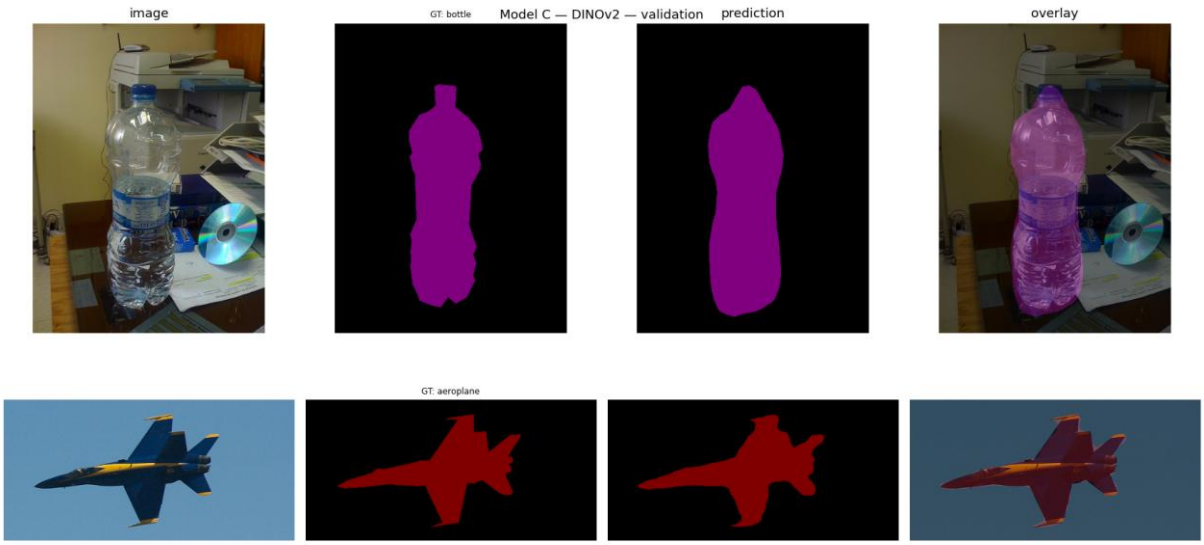
APPROACH

- **Classification:** two ResNet-18 models trained from scratch (about 3M and 11M parameters), and two models pre-trained on ImageNet-21k: ConvNeXt-Small and ViT-B/16. The pre-trained models were fine-tuned in two stages: first only the new head, then the whole network. All models used Asymmetric Loss, which reduces the weight of easy negative examples, and a multi-label stratified 80/20 split.

- **Segmentation:** U-Net and DeepLabV3+ (both with an EfficientNet-B4 encoder) as baselines, against a DINOv2 ViT-B/14 backbone kept frozen, with a small decoder (about 4M trainable parameters) on top. Training used a Dice + Cross-Entropy loss, strong augmentation, and test-time augmentation with two scales and flips.
- **Adversarial attacks:** white-box PGD attacks. On the classifier, an untargeted attack to hide the “person” class. On the segmenter, a targeted attack to turn every “person” pixel into “chair”.
- **Explainability:** Grad-CAM heatmaps show where the classifier looks before and after the attack.

RESULTS

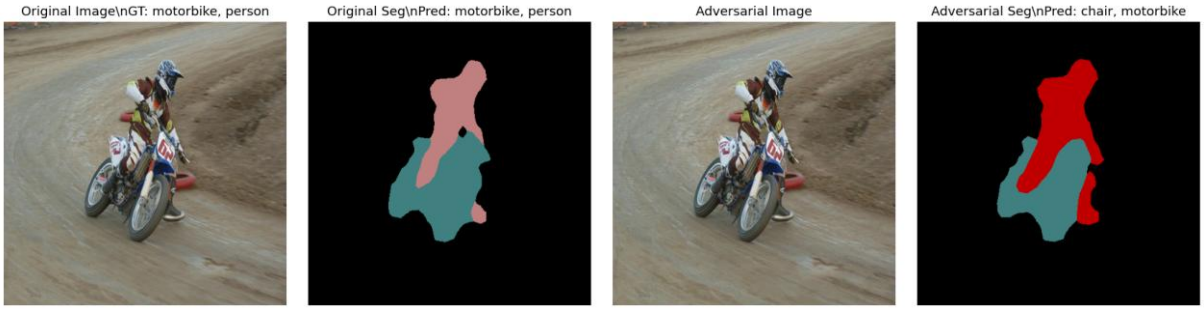
Task	Model	Validation score
Classification	A1: small ResNet-18 (from scratch)	mAP 0.401
Classification	A2: ResNet-18 (from scratch)	mAP 0.448
Classification	B1: ConvNeXt-Small (ImageNet-21k)	**mAP 0.907**
Classification	B2: ViT-B/16 (ImageNet-21k)	mAP 0.882
Segmentation	U-Net, EfficientNet-B4	mIoU 0.462
Segmentation	DeepLabV3+, EfficientNet-B4	mIoU 0.438
Segmentation	DINOv2 (frozen) + decoder	**mIoU ≈ 0.75**



DINOv2 segmentation on validation images: input, ground truth, prediction and overlay.

- **Pre-training matters most:** transfer learning more than doubled the classification score (about 0.45 to 0.91). More capacity without external data only added overfitting.
- **Self-supervised features win:** the frozen DINOv2 model beat the supervised baselines by about 28 mIoU points, while training only the small decoder on a single T4 GPU.
- **Where it struggles:** bicycle (IoU 0.28) and chair (0.45). Thin structures are smaller than the 14-pixel patch grid, and chairs are often unlabeled or hidden in cluttered scenes.

- **Attacks work very well:** the classifier attack lowered “person” confidence by 0.92 on average and fooled the model on every image. The person-to-chair attack on the segmenter succeeded on up to 97.5% of images (5 of 198 failed). The changed images still look normal to a human.



Person-to-chair attack: the rider is predicted as “chair” after a small, hard-to-see change.

CONCLUSIONS

At 749 images, the choice of pre-trained features matters more than the model design: ImageNet-21k transfer learning won in classification, and self-supervised DINOv2 won in segmentation. The attacks also showed that high accuracy does not mean robustness, so these models should not be trusted in security-critical uses without defenses such as adversarial training. Next steps would be a better split for rare classes, boundary refinement for thin objects, and testing black-box attacks.